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AIM : Data Visualization with Matplotlib: Data visualization on a dataset using matplotlib and seaborn libraries • Scatter plot • Line plot • Bar plot • Histogram • Box plot • Pair plot Theory : 1.What is the importance of data visualization in data analysis? How does visualization help in exploratory data analysis (EDA)? ANS : Data visualization is important because it helps present complex data in a simple, visual format like charts and graphs. This makes it easier to understand patterns, trends, and relationships. In Exploratory Data Analysis (EDA), visualization helps to:-Identify patterns and trends -Detect outliers or errors -Understand data distribution -Compare different variables -Make better decisions quickly 2.What is the difference between Matplotlib and other visualization libraries? ANS : Matplotlib is a basic and flexible plotting library in Python. It is widely used and gives full control over plots. Comparison: Matplotlib:-Low-level (more control) -Requires more code -Highly customizable Other libraries like Seaborn / Plotly: -High-level (easier to use) -Better default styles-More interactive (especially Plotly) 3.What is the purpose of the pyplot module in Matplotlib? ANS : The pyplot module provides a simple interface for creating plots, similar to MATLAB. It is used to:-Create figures and graphs -Plot data easily -Add labels, titles, and legends -Display or save plots Example: import matplotlib.pyplot as plt 4. Explain the process of creating a basic line plot. ANS : Steps: Import matplotlib.pyplot Define data (x and y values) Use plt.plot() to create the plot Add labels and title Display the plot using plt.show() Example: import matplotlib.pyplot as plt x = [1, 2, 3, 4] y = [10, 20, 25, 30] plt.plot(x, y) plt.xlabel("X-axis") plt.ylabel("Y-axis") plt.title("Basic Line Plot") plt.show() 5.Why is customization important in data visualization? What customization options are available in Matplotlib? ANS : Customization is important because it makes graphs:-More clear and readable -Visually appealing -Easy to understand for others Common customization options:-Titles (plt.title()) -Axis labels (plt.xlabel(), plt.ylabel()) -Colors and line styles-Markers -Legends (plt.legend()) -Grid (plt.grid()) -Figure size

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Generating dummy data for demonstration since 'cleaned_employee_data.csv' was not found.
data = {
    'Age': np.random.randint(22, 60, 50),
    'Salary': np.random.randint(30000, 120000, 50),
    'Department': np.random.choice(['HR', 'Engineering', 'Sales', 'Marketing', 'Finance'], 50)
}
df = pd.DataFrame(data)

print(df.head())
sns.set(style="whitegrid")

# 1. Scatter Plot
plt.figure()
plt.scatter(df['Age'], df['Salary'])
plt.xlabel('Age')
plt.ylabel('Salary')
plt.title('Scatter Plot (Age vs Salary)')
plt.show()

# 2. Line Plot
plt.figure(figsize=(4,3))
plt.plot(df['Age'], df['Salary'])
plt.xlabel('Age')
plt.ylabel('Salary')
plt.title('Line Plot (Age vs Salary)')
plt.show()

# 3. Bar Plot (Seaborn)
plt.figure(figsize=(4,3))
sns.barplot(x='Department', y='Salary', data=df)
plt.title('Bar Plot (Department vs Salary)')
plt.xticks(rotation=45)
plt.show()

# 4. Histogram
plt.figure(figsize=(4,3))
plt.hist(df['Salary'], bins=10)
plt.xlabel('Salary')
plt.ylabel('Frequency')
plt.title('Histogram of Salary')
plt.show()

# 5. Box Plot (Seaborn)
plt.figure(figsize=(4,3))
sns.boxplot(x='Department', y='Salary', data=df)
plt.title('Box Plot (Department vs Salary)')
plt.xticks(rotation=45)
plt.show()
```

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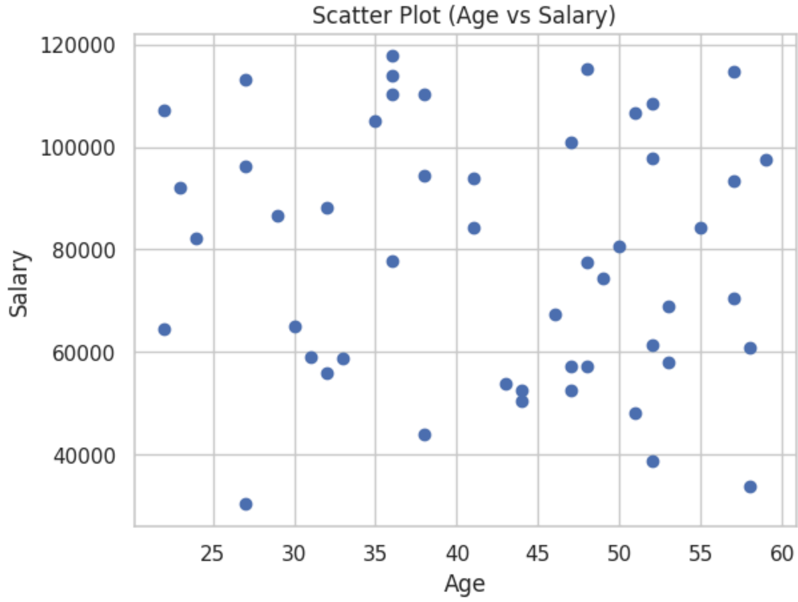
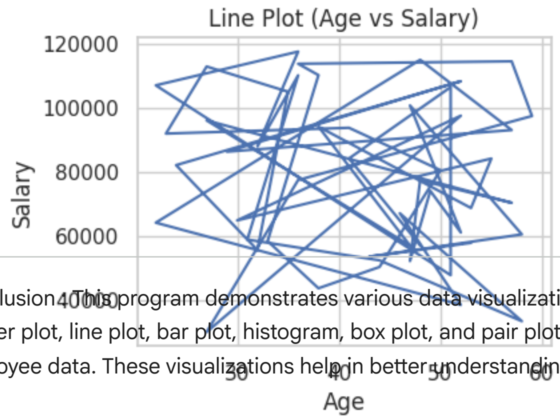
# 6. Pair Plot (Seaborn)
sns.pairplot(df,height=1.3)
plt.show()
```

	Age	Salary	Department
0	53	58082	HR
1	43	53771	Finance
2	58	60732	HR
3	47	100917	Sales
4	52	61311	Marketing





Conclusion: This program demonstrates various data visualization techniques using Matplotlib and Seaborn. Different plots such as scatter plot, line plot, bar plot, histogram, box plot, and pair plot are used to analyze relationships, distributions, and patterns in employee data. These visualizations help in better understanding of the dataset and support effective exploratory data analysis